

Chapter 2

Finding an Adequate Areal Unit to Map Crime: A Spatial Data Perspective



Rafael G. Ramos

Abstract Spatial analysis of crime often involves aggregating georeferenced crime data into areal units, generating a map of crime counts that can then be analyzed. Which areal unit should be used, though? This is an important decision, since changing the unit of area can impact analysis and results: the famous Modifiable Areal Unit Problem (Openshaw S, Taylor PJ, A million or so correlation coefficients: three experiments on the modifiable areal unit problem. In: Wrigley N (ed) *Statistical applications in the spatial sciences*, 127–144. Pion, London, 1979). In some cases, prior theoretical knowledge or even the research questions will recommend a specific areal unit (Weisburd D, Bruinsma GJ, Bernasco W, *Units of analysis in geographic criminology: historical development, critical issues, and open questions*. In: *Putting crime in its place*. Springer, New York, pp 3–31, 2009; Weisburd D, Groff ER, Yang SM, *The criminology of place: street segments and our understanding of the crime problem*. Oxford University Press, Oxford, 2012), but that might not always be the case. In this chapter, we present an approach that considers spatial properties of the original crime dataset to determine an adequate areal unit (see also Ramos RG, Silva BF, Clarke KC, Prates M, *J Quant Criminol* 37:419–443, 2021; Ramos 2019). Two properties are key: internal uniformity and robustness of the crime counts. If areas are too large, areal aggregation may hide crime clusters (low internal uniformity); if areas are too small, crime counts will be too prone to random fluctuations (low robustness of crime counts). For a more reliable crime map, these two factors need to be considered—the method presented in this chapter finds an areal unit that balances these two criteria. To illustrate the method, case studies using real-world crime data are shown at the end. A sensitivity analysis is also shown, illustrating how differences in some implementation details of the method affect the optimal unit but not to a radical extent, highlighting that the

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methodology identifies a range of acceptable granularities and the extremes that should be avoided.

Keywords Modifiable areal unit problem · Granularity · Error · Uniformity · GIS · Spatial analysis · Crime

2.1 Introduction

This chapter approaches the issue of choosing a unit of analysis for mapping crime i.e., its optimal granularity.¹ After a brief discussion on the nature of this issue, a methodology is presented that looks at the crime data itself for clues on how to determine the best unit. Case studies using real-world data of residential burglaries of two cities in Brazil are also shown here for illustration.

Mapping and geographical information systems are important tools for crime analysis and policing (Santos 2016; Chainey and Ratcliffe 2013; Wang 2012). Crime exhibits marked geographical patterns and understanding those can lead to a deeper understanding of crime itself. Crime has been found to frequently cluster in specific recurrent locations, often at specific recurrent times of the day, and knowing these patterns can reveal significant social and environmental determinants. This knowledge can also guide policy towards a more effective and efficient use of resources: *hotspot policing*, for instance, involves allocating patrols at very specific locations and times where crime has been observed to happen.

There are different ways to map crime. Sometimes, we are given less of a choice and the data is recorded in a pre-established unit, for instance, number of crimes per police district.² More and more, however, data for individual crime events is available, including their geolocation, given the analyst greater flexibility in how to process and represent the data. This, however, poses the problem of how to best map crime.

In this chapter, we focus on gridded choropleth maps created from a set of geo-located points, each point being a crime event. In this type of map, a uniform grid of square cells is projected over the study region, and crime events are aggregated per cell, creating a map of crime counts at different areas. Some other ways to map crime include non-gridded choropleth maps, with areal unit varying in shape and size (e.g., census tracts polygons); cluster maps, where a few major clusters are determined and highlighted from a set of crime points, and density mapping, where a smooth density surface is estimated from the point set, and often displayed as a heat map. Many of the topics discussed in this chapter are also relevant to non-gridded choropleth maps.

¹The terms ‘unit of analysis’, ‘areal unit’ and ‘granularity’ shall be used interchangeably here.

²In these cases in which the data is pre-aggregated to some areal unit, areal disaggregation methods could be an adequate avenue (Leyk et al. 2019; Goodchild and Lam 1980). That is not the focus of this discussion, however.

There are some advantages of using gridded choropleth maps. Choropleth maps are familiar and intuitive (though not perfect, see Becony   et al. 2022; Juergens 2020; Monmonier 2018) and areal crime counts can then be analyzed and modeled in a more straightforward manner against other areal datasets (e.g., census data, land-use). Per capita crime rates can also be estimated from areal maps, which can then be further analyzed. Moreover, many significant areal data products are gridded datasets, including population (Sims et al. 2023; Boo et al. 2022), poverty (CIESIN 2022), land-use/land-cover (Souza Jr et al. 2020) and cellphone tower (Unwired Labs 2023) datasets. Of course, certain contexts may hamper the use of gridded choropleth maps: pertinent dataset (e.g. demographic and socioeconomic) might only be available in non-gridded formats³ (e.g. census units, political units) and/or practical considerations may require the use of other units (e.g. evaluation of police agencies might require metrics to be aggregated at the district level). Nevertheless, gridded choropleth maps and datasets have not only various advantages but are progressively becoming more available.

One key issue for mapping crime in a grid (or in any other type of areal unit) is determining the unit of analysis i.e., its granularity. Changing the unit of analysis can change your results, what is known as the Modifiable Areal Unit Problem (Openshaw and Taylor 1979; O’Sullivan and Unwin 2014, p. 30). A well-known example of MAUP in practice is gerrymandering, in which changes in district boundaries can affect the outcome of district-based electoral systems. Another related issue is the Ecological Fallacy (Robinson 1950; O’Sullivan and Unwin 2014, p. 30), in which correlation at broader units are incorrectly assigned to finer ones.

In this chapter, I propose a method that looks at the data’s own patterns to find an appropriate unit for mapping crime in a grid. Namely, the method here presented analyses the data and finds a unit of analysis that is not too broad as to conceal many significant clusters, but not too fine that the crime counts become unstable—i.e. a compromise between internal uniformity and robustness. Section 2.2 discusses some significant background to the problem of granularity in crime mapping, and the method is described in detail in Sect. 2.3. In Sect. 2.4, an application of the method is shown, including a sensitivity analysis of the results, with Sect. 2.5 wrapping up this chapter with a summary and a discussion of caveats and limitations for the methodology.

2.2 Historical Background

The issue of choosing an adequate area unit for geographical analysis of crime is now well known—the aforementioned MAUP (Openshaw and Taylor 1979) and Ecological Fallacy (Robinson 1950; O’Sullivan and Unwin 2014, p. 30) issues. A

³Again, areal disaggregation methods may provide a solution to this issue, although they introduce their own set of assumptions in resampling the data.

more detailed account of previous studies can be found in Wong (2009), and here the contributions most related to this chapter are highlighted.

The effects introduced by areal aggregation have been known at least since Gehlke & Biehl (1934), with the terms Ecological Fallacy and MAUP being coined, respectively, by Robinson (1950) and Openshaw and Taylor (1979). Preceding even the coinage of the term MAUP, there were attempts at solving the problem of choosing an areal unit: for instance, Moellering and Tobler (1972) proposed an analysis of variance method to choose an aggregation unit; Batty (1976) proposed using spatial entropy, and Openshaw (1977) argued that the optimal geographical division is that which leads to the optimal fitness of a model. Of notice, neither of these approaches explicitly considered spatial uniformity and/or robustness in the choice of areal unit, two key criteria of the methodology presented in this chapter. Spatial uniformity (or rather its opposite, spatial clustering) is considered by Openshaw et al. (1987)'s Geographic Analysis Machine as a criterion in processing point data, although the spatial bandwidth had to be pre-specified by the analyst and the final product was not a gridded map but a map of hotspots; also importantly, King (1997) elaborated on how considering internal uniformity of attributes in areal aggregation can bypass the problems of the Ecological Fallacy, highlighting the connection between the two problems. Similar considerations on internal uniformity, MAUP and the Ecological Fallacy can be seen in Holt et al. (1996) and Tranmer and Steel (1998). Finally, Sun and Wong (2017) and Spielman and Folch (2015) proposed methods for aggregating finer units into broader ones as a strategy to improve attribute robustness, with the latter method also considering attribute similarity when combining areas (i.e. prioritizing broad uniform units), thus anticipating some aspects of the method here described. These methods, however, do not consider point data such as geocoded crime reports, topic of this chapter.

For most of its history, however, the question of 'what is an adequate geographical unit' was not in the forefront of spatial criminology, becoming a topic of investigation only in the last decades (Weisburd et al. 2009). Up until the 1980s, most notable studies and theories in Criminology that included a spatial component tended to employ meso and/or macro units of analysis: neighborhoods, districts, cities, countries, etc. For instance, in the earliest studies on the geography of crime, both Guerry (1833) and Quetelet (1842) analyzed crime rates in France per department; in Social Disorganization theory (Shaw and McKay 1942), neighborhoods and their relation to the city are the fundamental unit determining crime, and in Cohen and Felson (1979), the seminal paper for the Routine Activities theory, longitudinal data aggregated for the whole of the United States was used, despite the theory's spatial components.

Starting with the work of Sherman et al. (1989), however, criminologists observed that crime was highly concentrated at very fine units. For instance, "just over half (50.4%) of all calls to the police for which cars were dispatched" going "to a mere 3.3% of all addresses and intersections" (Sherman et al. 1989). This type of concentration found parallel in other social phenomena—i.e., Zipf's law and the Pareto distribution (Newman 2005); furthermore, this clustering was accompanied by seasonal and/or daily patterns, and these patterns (spatial and temporal) were observed

to be persistent (Weisburd et al. 2004, 2009, 2012; Groff et al. 2009; Braga et al. 2011; Curman et al. 2015). These observations have motivated further research using micro units and have bolstered strategies such as *hotspot policing*, in which police patrols and other resources are allocated at specific places and times where crime is observed to concentrate. Culminating with the Law of Crime Concentration (Weisburd et al. 2014).

In tandem, theoretical frameworks on crime and geography also zoomed into finer levels. Crime pattern theory (Brantingham and Brantingham 1984, 1993) proposed how specific activity nodes and connecting paths could explain the geography of crime at intra-urban levels—sometimes intra-neighborhood—and criminology of place (Weisburd et al. 2012) emphasized the nature of specific facilities in determining crime occurrences at very specific locations. Advances in geoinformation science and technologies can also be credited in allowing this finer type of data analysis to be implemented (Chainey and Ratcliffe 2013; Anselin et al. 2000).

Empirical evidence and theory have converged to the idea that micro-units are key to understand spatial patterns of crime (Weisburd et al. 2009, 2012). Oft times, these micro-units have been street segments; in parallel, some have argued that “finest is best” (Oberwittler and Wikström 2009).

Some recent studies (Alazawi et al. 2022; Cheng et al. 2022; Drake et al. 2022; Wheeler 2021; Buil-Gil et al. 2021; Ramos et al. 2021; Malleson et al. 2019), however, have raised some cautionary notes to this approach. Rather than discourage the use of micro-units, these studies point out that some units may at times be too fine, and that some additional forethought, nuance and compromise is required to map crime with micro-units⁴—a principle also supported by current knowledge on MAUP and the Ecological Fallacy. This chapter should be read in the same vein.

2.3 Methodology

2.3.1 Conceptual Overview

The method here proposed looks at patterns in the crime data itself for cues to determine what is an adequate granularity for that dataset. The method addresses two key concerns: the granularity should not be too broad that it conceals significant clusters of crime, and it should not be too fine that the calculated crime counts are too susceptible to error.

When a granularity is too broad, crime may not be evenly distributed within each unit. For instance, in one cell, there may be subareas in which crime is much higher and others in which it is nearly absent—details that will be lost in aggregating over the broader cell. This can lead to inference problems such as the previously discussed Ecological Fallacy (O’Sullivan and Unwin 2014, p. 30). This evenness (or

⁴This chapter develops over the work presented in Ramos et al. (2021).

not) of distribution of crime within cells shall be referred here as internal uniformity.⁵ In the previous example, we should say then that the granularity is too broad because it has low internal uniformity. In Sect. 2.3.3, methods for estimating internal uniformity are described.

Naturally, a solution to granularities that are too broad is to take finer ones. If a granularity is too fine, however, the resulting crime counts per cell may be too unstable and prone to error. Consider the following analogy: one basketball player attempts two throws and scores once, while a second player makes 100 attempts and scores 25 times. The first player has a success rate of 50%, twice that of the second player. However, one would hardly consider the first measurement for rate of success to be as accurate as the second. Similarly, the finer the granularity, the larger the number of cells for counting the same number of crime points. With too many cells, a great proportion might even not contain any crimes at all.

Crime count maps are typically interpreted also as a map of probability of criminal occurrence. This is further reinforced by the observation that crime clusters tend to be closely connected to their place of occurrence (Weisburd et al. 2012). However, crime count maps (as well as the originating crime point dataset) are but finite realizations of this assumed underlying probability distribution. As with any probabilistic event, some natural variation in the point distribution (and associated counts) can be expected. Moreover, as a real-world dataset, other sources of error and/or inaccuracies can also occur, such as errors or ambiguities in how crime is recorded, geocoding errors, and underreporting (Buil-Gil et al. 2021; Chainey and Ratcliffe 2013, chapter 3). How reliable the crime counts are in face of these fluctuations and sources of error shall be referred here as robustness to error. In Sect. 2.3.4, methods for estimating robustness to error are described.

As can be seen, both internal uniformity and robustness to error are important for making reliable crime maps, and a compromise between the two is necessary, avoiding granularities that are either too broad or too fine. In the following subsections, I describe how this can be implemented.

2.3.2 *Summary of the Method in Three Steps*

The approach presented here for determining an adequate granularity to map crime looks at properties from the crime data itself, and can be summarized in three steps:

1. Estimate the internal uniformity for different granularities.
2. Estimate the robustness to error for different granularities.
3. Find a granularity that offers a compromise between internal uniformity and robustness to error.

These steps are detailed in the following subsections.

⁵Bunge (1966) lists uniformity as one of the key criteria for defining regions.

2.3.3 Estimating Internal Uniformity

Internal uniformity is estimated through a series of complete spatial randomness (CSR) tests. Given a granularity g , a set of n quadrats of that granularity $S_g = \{s_{1g}, s_{2g}, \dots, s_{ng}\}$ is generated over the domain of crime points P being tested, and then for each quadrat s_{ig} , a spatial randomness test is performed for the points located inside it. The proportion of samples in S_g that fail to reject the complete spatial randomness hypothesis at significance α is calculated and attributed as the internal uniformity level u_g of the point set P at granularity g . This process is repeated for a range of different granularities. Figures 2.1 and 2.2 illustrate this process.

There are different manners that one can draw these samples, and different methods for testing complete spatial randomness. Two methods for generating samples are considered here, as well as two methods for testing spatial randomness, leading to four different combinations of metrics, which can then be compared.

For generating the samples, the first approach determines a set of r random quadrats of the same granularity (size) inside the bounding rectangle that delimits the point set P . The second approach divides the bounding rectangle into a set of contiguous quadrats of the same granularity, as in standard quadrat analysis. Figure 2.3 illustrates these two different approaches. The contiguous samples approach has the potential advantage of always covering the entire study area and not requiring the analyst to decide on a sample size r . In practice, the random samples approach tends to be computationally faster and yields similar results (as seen in Sect. 2.5); nevertheless, the completeness of the contiguous samples approach is useful as a baseline to evaluate the random samples variation.

As for the spatial randomness test used, the first approach is to use a nearest neighbor distance test—here, the version of the test employed is the Clark-Evans test (O’Sullivan and Unwin 2014). The second approach is through a quadrat count test, as shown by O’Sullivan and Unwin (2014).

Table 2.1 synthesizes the four combinations employed in this study for estimating internal uniformity.

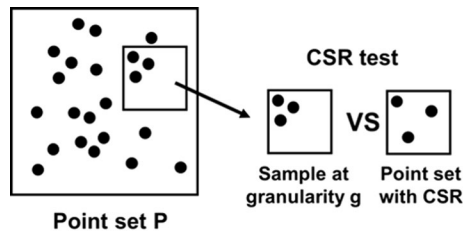


Fig. 2.1 Evaluating the spatial randomness of one sample taken at granularity g . The test is repeated for multiple samples at the same granularity over the point set, and the proportion of points that fail to reject the CSR hypothesis is assigned as the internal uniformity

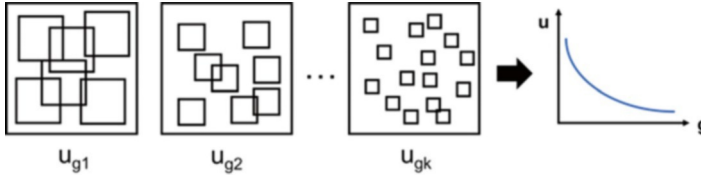
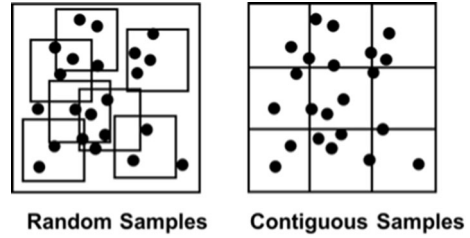


Fig. 2.2 Internal Uniformity is calculated for a range of different granularities

Table 2.1 The four metrics (A, B, C and D) proposed for estimating internal uniformity of a point set, varying according to the CSR test employed and the sampling procedure

CSR test	Sampling procedure	
	Random samples	Contiguous samples
Nearest neighbor	A	B
Quadrat count	C	D

Fig. 2.3 Two approaches for collecting samples to be tested for spatial randomness



2.3.4 Estimating Robustness to Error

Robustness of the measured crime counts is evaluated in relation to the expected proportional variation of crime incidence (i.e., the estimated coefficient of variation of the observed crime incidence, see Krishnamoorthy 2016 for reference). For example, a crime rate of 100 crimes per year with a coefficient of variation of 5% is a more stable measure than a rate of 100 crimes per year with a 20% coefficient of variation.⁶ As the coefficient of variation is a positive quantity, we define a metric r_g for robustness to error at granularity g as the exponential of a constant times the average coefficient of variation for a sample $S_g = \{s_{1g} s_{2g} \dots s_{ng}\}$ of n quadrats of granularity g as in Eq. 2.1.

$$r_g = \exp \left(-k \times \frac{1}{n} \sum_{i=1}^n \text{coef_var}(s_{ig}) \right) \quad (2.1)$$

⁶The coefficient of variation is here considered a more adequate metric to robustness than other metrics like standard deviation or variance, since the main interest is not in measuring simply the potential variation of crime counts, but of how much it should impact the rate of crime observed. Similarly, related metrics like variance-to-mean ratio are not as adequate in that the scales for the units are not the same, the unit of variance being the square of the unit of the mean. See Brown (1998) for a similar example.

The exponential is applied to assure that r_g will vary between (0, 1) and, therefore, be comparable to the uniformity metric. For the choice of k , we suggest $k = 3$, since at that value a coefficient of variation of 1 will yield a robustness close to zero (0.05), while a coefficient of variation of 0 will yield a maximum robustness of 1. Although the behavior of the robustness metric itself is in part dependent on the specific value of k chosen, we show in Appendix A how our estimate of the best granularity is not sensitive to the value of k .

To estimate the coefficient of variation for the observed crime distributions, three different methods were compared; additionally, two different approaches for selecting the samples were employed (similar to the calculation of internal uniformity).

The first method for estimating the coefficient of variation, named here *binomial estimation*, begins with comparing the number of crime events k_i within a quadrat of granularity g to the total number k of observed crime events. By assuming that the ratio $p_i = k_i/k$ is a good approximation of the likelihood of a crime event being located inside a quadrat s_{ig} , the coefficient of variation for that sample follows that of a binomial distribution, which is given by Eq. 2.2.

$$\text{coef_var}(s_{ig}) = p_i(1 - p_i) \frac{1}{\sqrt{k}} \quad (2.2)$$

The second method, called here *Poisson estimation*, assumes that the crime incidence within a quadrat of granularity g follows a Poisson distribution and then proceeds to fit the observed sample to that distribution. Since only one sample is used in the fitting, the estimated coefficient of variation for the Poisson distribution is given by Eq. 2.3:

$$\text{coef_var}(s_{ig}) = \frac{1}{\sqrt{k_i}} \quad (2.3)$$

The rationale behind the second method is that for sufficiently large number events the binomial distribution can be approximated by the Poisson distribution.

Finally, the third and last method, called here *resampled estimation*, effectively combines the first two for estimating the coefficient of variation. In the Poisson distribution, the coefficient of variation is dependent on the lambda parameter. However, not only the estimated lambda is uncertain, but its relation to the coefficient of variation is non-linear. Thus, to estimate the coefficient of variation while taking account of the uncertainty in the parameter lambda, a resampling approach was used: m samples of crime counts are simulated following a binomial distribution with probability $p_i = k_i/k$ for k events, and for each sample, the coefficient of variation is estimated by fitting it to a Poisson distribution. Finally, the simulated coefficients of variation are averaged to estimate the expected value of the coefficient of variation:

$$\text{coef_var}(s_{ig}) = \frac{1}{m} \sum_{j=1}^m \frac{1}{\sqrt{\lambda_{ij}}} \quad (2.4)$$

$$s.t. \lambda_{ij} = \text{Binom}\left(\frac{k_i}{k}, k\right)$$

2.3.5 *Finding a Balance Between Internal Uniformity and Robustness to Error*

Having computed the metrics for internal uniformity and robustness to error, the final step is to determine a granularity that offers a compromise between each metric. Here, three approaches are proposed (and later compared) for estimating a granularity that provides a compromise between uniformity and robustness, but it is worth noting that different approaches (and definitions of balance) for finding this compromise are possible (Miettinen 2012; Cho et al. 2017). Additionally, the term “best granularity” should be understood under this restricted sense of providing a balance between uniformity and robustness (of which the following criteria are three of many possible) and avoiding extremes of low uniformity or low robustness.

The first approach is by fitting a curve for internal uniformity varying as a function of robustness to error, and then finding the point where its derivative is equal to minus one:

$$\frac{du(g_{\text{best}})}{dr(g_{\text{best}})} = -1 \quad (2.5)$$

The rationale is that, since both metrics are percentile values, a derivative equal to minus one means that any marginal gain in robustness due to a small change in granularity will lead to a loss in uniformity of the same magnitude, and vice-versa.

The second approach proposed here is to find the granularity that maximizes the product of internal uniformity and robustness to error:

$$\max u(g)r(g) \quad (2.6)$$

The motivation for using this criterion is that it requires both metrics to score high in order to yield a high value: for instance, regardless of the internal uniformity, if the robustness to error is zero, the product will be zero. Furthermore, since both metrics are percentiles, there is no need to standardize their values, and the magnitudes can be compared in a straightforward way.

The third and final criterion proposed is finding the granularity that maximizes the sum of internal uniformity with robustness to error:

$$\max(u(g) + r(g)) \quad (2.7)$$

In contrast to the previous criteria, this privileges granularities that offer either a good internal uniformity or a good robustness to error, but there is less of a penalty if one of the metrics is relatively low.

Since the metrics for internal uniformity and robustness to error are estimated for a finite set of granularities, it is recommend fitting a spline curve (or equivalent) to each metric and then applying the desired criterion for finding the balance. In this manner, outliers in the estimates can be better avoided.

Multiple criteria can be tested in parallel. As will be shown in the sensitivity analysis (Sect. 2.4.3), changing the criterion changes the optimal unit, but the difference is not usually radical. Instead, it reveals that there is a range of acceptable granularities that can be considered, and extremes to avoid.

2.4 Case Study

To illustrate the methodology, a case study using two different datasets is shown here. The method is applied to residential burglaries in two large cities in Brazil—Belo Horizonte (2008–2014) and Campinas (2010–2013). We use random sampling⁷ with a nearest neighbor method for estimating the internal uniformity (method A in Table 2.1, Poisson estimation for robustness to error (Eq. 2.3), and the product criterion for finding the balance (Eq. 2.6). As will be shown, the estimated optimal granularities will differ between the two cases. Finally, a sensitivity analysis for the case of Belo Horizonte is conducted, to show how the variations proposed in the method tend to estimate similar optimal granularities, given the same dataset.

2.4.1 Data

The data used in this case study derives from crimes reported to the police (in this case, residential burglaries), which in Brazil are registered as *boletins de ocorrência*. These reports typically contain the address where the crime took place, allowing these *boletins de ocorrência* to be converted into a set of georeferenced points ready to be processed. The residential burglary data for Belo Horizonte comprises 44,560 georeferenced points and was provided by the Military Police of the State of Minas Gerais; the residential burglary data for Campinas comprises 4750 georeferenced points and was provided by the Civil Police of the State of São Paulo, being

⁷Note that when using random sampling, some small variations can be expected if the method is to be repeated on the same dataset. However, random sampling yields similar results to contiguous sampling (as will be shown in Sect. 2.4.3) and is generally faster.

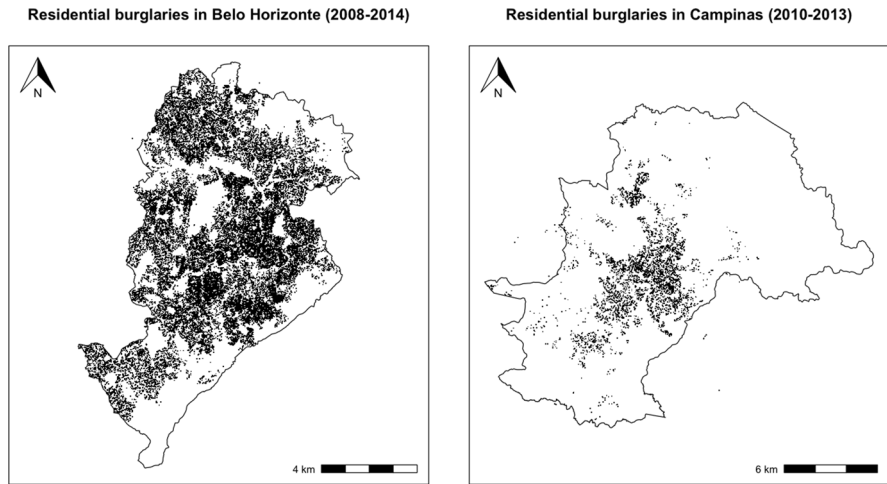


Fig. 2.4 Point maps of the residential burglaries for Belo Horizonte (2008–2014) and Campinas (2010–2013)

collected from the Criminal Information System (Infocrim). These two datasets are mapped in Fig. 2.4.

2.4.2 Results

The optimal granularity estimated by the proposed method is a grid cell of 325 m on a side for the case of residential burglaries in Belo Horizonte, and 725 m for Campinas. In the case of Belo Horizonte, the optimal grid size corresponds to an internal uniformity of 0.60 and a robustness to error of 0.38, indicating that 60% of the cells contain no significant clusters within them and that the average coefficient of variation expected for crime counts is of 32%. For the case of Campinas, the internal uniformity is 0.78 and the robustness 0.21, corresponding to 78% of the cells showing complete spatial randomness internally and an average coefficient of variation of 52% for the crime counts.

Figures 2.5 and 2.6 show the internal uniformity and the robustness to error varying with granularity in both cases, as well as the optimal granularity estimated by the product criterion.

Finally, maps of crime counts using the estimated optimal granularities are shown in Fig. 2.7.

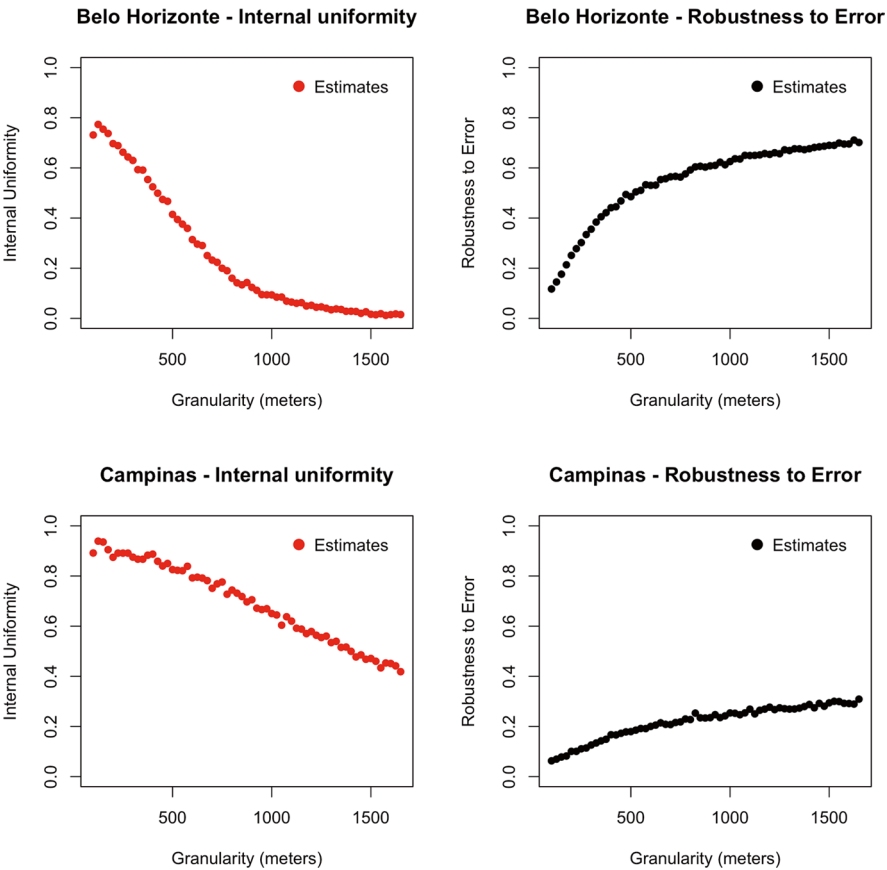


Fig. 2.5 Internal uniformity and robustness to error varying with granularity, for the cases of Belo Horizonte and Campinas

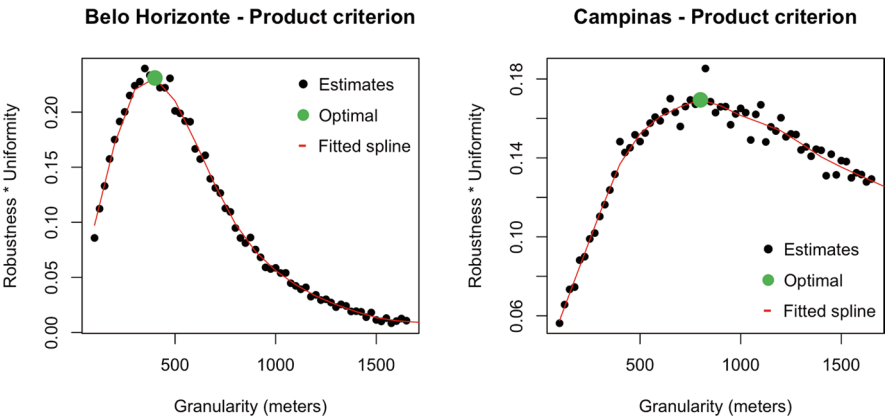


Fig. 2.6 Optimal granularity for Belo Horizonte (325 m) and Campinas (725 m), using the product criterion

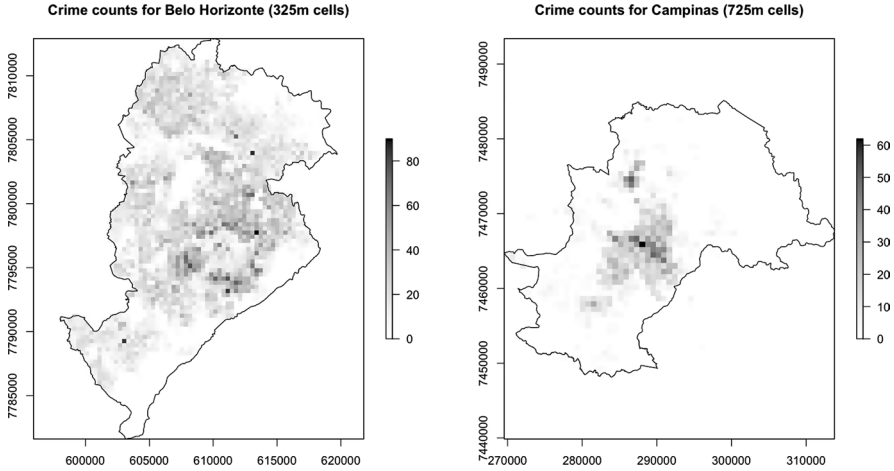


Fig. 2.7 Crime count maps using the estimated optimal granularities for Belo Horizonte (325 m cells) and Campinas (725 m cells)

2.4.3 Sensitivity Analysis

In Fig. 2.8, internal uniformity and robustness to error for residential burglaries in Belo Horizonte is estimated according to the different methods described in Sects. 2.3.3 and 2.3.4. As illustrated in the picture, the estimates vary little when different methods are used.

In Fig. 2.9, the optimal granularity is determined using the three different criteria described in Sect. 2.3.5. The estimated optimal granularities according to each criterion are 225 m, 325 m, and 325 m, for balance of gains, product, and sum criteria, respectively.

Finally (Fig. 2.10), the optimal granularity estimates are compared when using three different values for the parameter k (as in Eq. 2.1) for the robustness to error calculation. As can be seen, while the different parameter affects the metric, it does not radically affect the tradeoff shape nor the estimate in the end.

In summary, variations in the implementation of the methodology do change the optimal unit, but the difference is not radical, instead revealing a range of acceptable granularities and the extremes to avoid.

2.5 Conclusions

In this chapter, an approach was proposed for finding an adequate unit of analysis to map crime counts. In this approach, the data that we desire to map is analyzed to determine what is the best granularity based on two criteria: internal uniformity and robustness to error. The granularity cannot be too fine that the crime counts are too

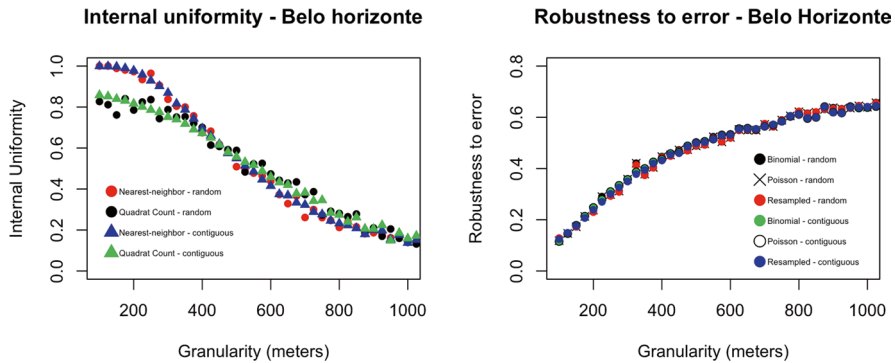


Fig. 2.8 Internal uniformity and robustness to error estimated according to the different methods described in Sects. 2.3.3 and 2.3.4

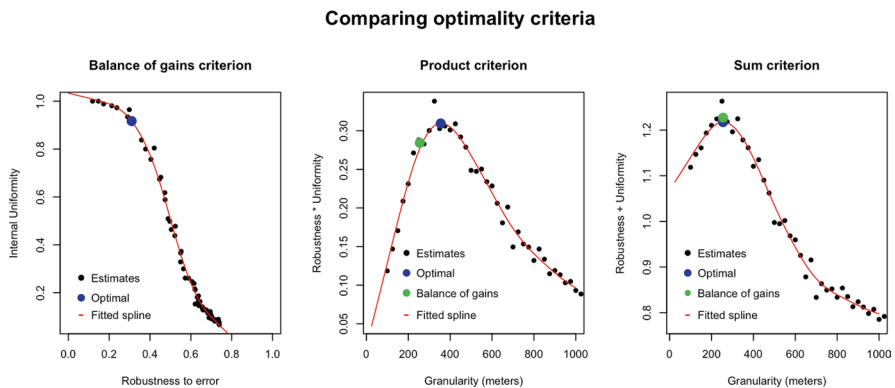


Fig. 2.9 Optimal granularities estimated through three different criteria

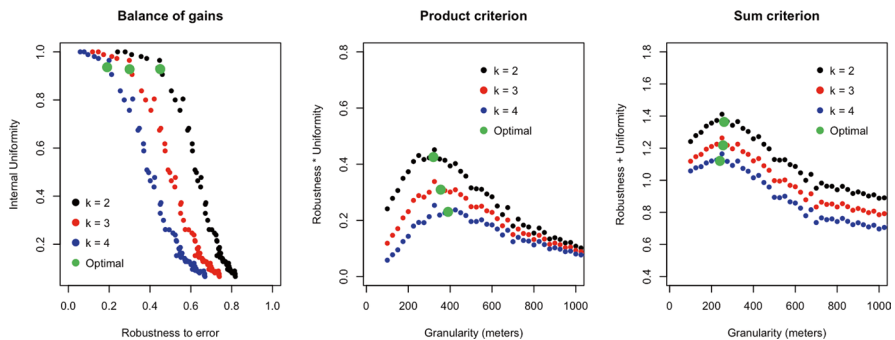


Fig. 2.10 Robustness to error estimated using different values for k

unstable and prone to error (poor robustness to error), nor too broad that it conceals many significant clusters (poor internal uniformity). The method here presented estimates the internal uniformity and the robustness to error for different granularities, then determining a granularity that offers a good compromise between the two criteria.

The method was illustrated using residential burglary data for two cities in Brazil—Belo Horizonte and Campinas. The method yielded different recommended granularities for each case, recommending a finer unit for the case of Belo Horizonte when compared to Campinas. The dataset for Belo Horizonte contains a larger volume of points than Campinas' (around ten times the amount), and thus it is expected that the recommended granularity would be finer, since with a greater volume of data the crime counts tend to be more stable. However, this may also vary with other patterns in the data, such as the degree of clustering.

Finally, the sensitivity tests showed that varying the parameters and other implementation details in the methodology can alter the optimal granularity, but not to a radical degree. As such, this method should not be thought of as providing one single best granularity based on a balance of internal uniformity and robustness to error, but that as providing a range of acceptable granularities and extremes that should be avoided. This interpretation to the proposed method allows it to be combined to other criteria for determining an adequate unit, such as policy and theoretical considerations in criminology.

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